5.4 Independence of Path

INTRODUCTION In Section 5.1 we saw that when a real function f possesses an elementary antiderivative, that is, a function F for which $F'(x) = f(x)$, a definite integral can be evaluated by the fundamental theorem of calculus:

$$\int_a^b f(x)\,dx = F(b) - F(a). \tag{1}$$

Note that $\int_a^b f(x)\,dx$ depends only on the numbers a and b at the initial and terminal points of the interval of integration. In contrast, the value of a real line integral $\int_C P\,dx + Q\,dy$ generally depends on the curve C. However, there exist integrals $\int_C P\,dx + Q\,dy$ whose value depends only on the initial point A and terminal point B of the curve C, and not on C itself. In this case we say that the line integral is **independent of the path**. For example, $\int_C y\,dx + x\,dy$ is independent of the path. See Problems 19–22 in Exercises 5.1. A line integral that is independent of the path can be evaluated in a manner similar to (1). It seems natural then to ask:

Is there a complex version of the fundamental theorem of calculus?
Can a contour integral $\int_C f(z)\,dz$ be independent of the path?

In this section we will see that the answer to both of these questions is yes.

Path Independence

The definition of **path independence** for a contour integral $\int_C f(z)\,dz$ is essentially the same as for a real line integral $\int_C P\,dx + Q\,dy$.

DEFINITION 5.4.1 Independence of the Path

Let z_0 and z_1 be points in a domain D. A contour integral $\int_C f(z)\,dz$ is said to be **independent of the path** if its value is the same for all contours C in D with initial point z_0 and terminal point z_1.

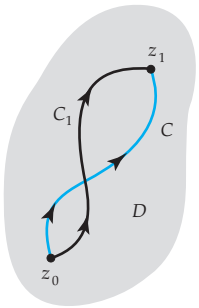


FIGURE 5.4.1 If f is analytic in D, integrals on C and C_1 are equal.

At the end of the preceding section we noted that the Cauchy-Goursat theorem also holds for closed contours, not just simple closed contours, in a simply connected domain D. Now suppose, as shown in FIGURE 5.4.1, that C and C_1 are two contours lying entirely in a simply connected domain D and both with initial point z_0 and terminal point z_1. Note that C joined with the opposite curve $-C_1$ forms a closed contour. Thus, if f is analytic in D, it follows from the Cauchy-Goursat theorem that

$$\int_C f(z)\,dz + \int_{-C_1} f(z)\,dz = 0. \tag{2}$$

But (2) is equivalent to

$$\int_C f(z)\,dz = \int_{C_1} f(z)\,dz. \tag{3}$$

The result in (3) is also an example of the principle of deformation of contours introduced in (5) of Section 5.3. We summarize the last result as a theorem.

THEOREM 5.4.1 Analyticity Implies Path Independence

Suppose that a function f is analytic in a simply connected domain D and C is any contour in D. Then $\int_C f(z)\,dz$ is independent of the path C.

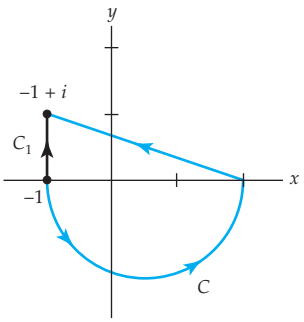


FIGURE 5.4.2 Contour for Example 1

EXAMPLE 1 Choosing a Different Path

Evaluate $\int_C 2z\,dz$, where C is the contour shown in color in FIGURE 5.4.2.

Solution Since the function $f(z) = 2z$ is entire, we can, in view of Theorem 5.4.1, replace the piecewise smooth path C by any convenient contour C_1 joining $z_0 = -1$ and $z_1 = -1 + i$. Specifically, if we choose the contour C_1 to be the vertical line segment $x = -1$, $0 \le y \le 1$, shown in black in Figure 5.4.2, then $z = -1 + iy$ and $dz = i\,dy$. Therefore,

$$\int_C 2z\,dz = \int_{C_1} 2z\,dz = -2\int_0^1 y\,dy - 2i\int_0^1 dy = -1 - 2i. \qquad \equiv$$

A contour integral $\int_C f(z)\,dz$ that is independent of the path C is usually written $\int_{z_0}^{z_1} f(z)\,dz$, where z_0 and z_1 are the initial and terminal points of C. Hence, in Example 1 we can write $\int_{-1}^{-1+i} 2z\,dz$.

There is an easier way to evaluate the contour integral in Example 1, but before proceeding we need another definition.

DEFINITION 5.4.2 Antiderivative

Suppose that a function f is continuous on a domain D. If there exists a function F such that $F'(z) = f(z)$ for each z in D, then F is called an **antiderivative** of f.

For example, the function $F(z) = -\cos z$ is an antiderivative of $f(z) = \sin z$ since $F'(z) = \sin z$. As in calculus of a real variable, the most general antiderivative, or **indefinite integral,** of a function $f(z)$ is written $\int f(z)\,dz = F(z) + C$, where $F'(z) = f(z)$ and C is some complex constant. For example, $\int \sin z\,dz = -\cos z + C$.

Since an antiderivative F of a function f has a derivative at each point in a domain D, it is necessarily analytic and hence continuous at each point in D. We are now in a position to prove the complex analogue of (1).

Recall, differentiability implies continuity. ▶

THEOREM 5.4.2 Fundamental Theorem for Contour Integrals

Suppose that a function f is continuous on a domain D and F is an antiderivative of f in D. Then for any contour C in D with initial point z_0 and terminal point z_1,

$$\int_C f(z)\,dz = F(z_1) - F(z_0). \tag{4}$$

Proof We will prove (4) in the case when C is a smooth curve parametrized by $z = z(t)$, $a \le t \le b$. The initial and terminal points on C are then $z(a) = z_0$ and $z(b) = z_1$. Using (11) of Section 5.2 and the fact that $F'(z) = f(z)$ for each z in D, we then have

$$\begin{aligned}\int_C f(z)\,dz &= \int_a^b f(z(t))z'(t)\,dt = \int_a^b F'(z(t))z'(t)\,dt \\ &= \int_a^b \frac{d}{dt}F(z(t))\,dt \quad \leftarrow \text{chain rule} \\ &= F(z(t))\,|_a^b \\ &= F(z(b)) - F(z(a)) = F(z_1) - F(z_0).\end{aligned}$$

$\equiv$

EXAMPLE 2 Applying Theorem 5.4.2

In Example 1 we saw that the integral $\int_C 2z\,dz$, where C is shown in Figure 5.4.2, is independent of the path. Now since the $f(z) = 2z$ is an entire function, it is continuous. Moreover, $F(z) = z^2$ is an antiderivative of f since $F'(z) = 2z = f(z)$. Hence, by (4) of Theorem 5.4.2 we have

$$\int_{-1}^{-1+i} 2z\,dz = z^2\Big|_{-1}^{-1+i} = (-1+i)^2 - (-1)^2 = -1 - 2i.$$

EXAMPLE 3 Applying Theorem 5.4.2

Evaluate $\int_C \cos z\,dz$, where C is any contour with initial point $z_0 = 0$ and terminal point $z_1 = 2 + i$.

Solution $F(z) = \sin z$ is an antiderivative of $f(z) = \cos z$ since $F'(z) = \cos z = f(z)$. Therefore, from (4) we have

$$\int_C \cos z\,dz = \int_0^{2+i} \cos z\,dz = \sin z\,|_0^{2+i} = \sin(2+i) - \sin 0 = \sin(2+i).$$

If we desire a complex number of the form $a + ib$ for an answer, we can use $\sin(2 + i) \approx 1.4031 - 0.4891i$ (see part (b) of Example 1 in Section 4.3). Hence,

$$\int_C \cos z\,dz = \int_0^{2+i} \cos z\,dz \approx 1.4031 - 0.4891i.$$

Some Conclusions We can draw several immediate conclusions from Theorem 5.4.2. First, observe that if the contour C is closed, then $z_0 = z_1$ and, consequently,

$$\oint_C f(z)\,dz = 0. \tag{5}$$

Next, since the value of $\int_C f(z)\,dz$ depends only on the points z_0 and z_1, this value is the same for any contour C in D connecting these points. In other words:

- *If a continuous function f has an antiderivative F in D, then $\int_C f(z)\,dz$ is independent of the path.* (6)

In addition, we have the following sufficient condition for the existence of an antiderivative.

- *If f is continuous and $\int_C f(z)\,dz$ is independent of the path C in a domain D, then f has an antiderivative everywhere in D.* (7)

The last statement is important and deserves a proof.

Proof of (7) Assume f is continuous and $\int_C f(z)\,dz$ is independent of the path in a domain D and that F is a function defined by $F(z) = \int_{z_0}^{z} f(s)\,ds$, where s denotes

a complex variable, z_0 is a fixed point in D, and z represents any point in D. We wish to show that $F'(z) = f(z)$, that is,

Important ▶

$$F(z) = \int_{z_0}^{z} f(s)\,ds \tag{8}$$

is an antiderivative of f in D. Now,

$$F(z+\Delta z) - F(z) = \int_{z_0}^{z+\Delta z} f(s)\,ds - \int_{z_0}^{z} f(s)\,ds = \int_{z}^{z+\Delta z} f(s)\,ds. \tag{9}$$

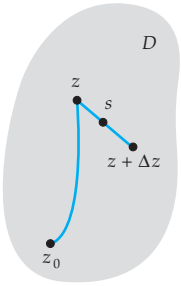


FIGURE 5.4.3 Contour used in proof of (7)

Because D is a domain, we can choose Δz so that $z + \Delta z$ is in D. Moreover, z and $z + \Delta z$ can be joined by a straight segment as shown in FIGURE 5.4.3. This is the contour we use in the last integral in (9). With z fixed, we can write

$$f(z)\Delta z = f(z)\int_{z}^{z+\Delta z} ds = \int_{z}^{z+\Delta z} f(z)\,ds$$

or

$$f(z) = \frac{1}{\Delta z}\int_{z}^{z+\Delta z} f(z)\,ds. \tag{10}$$

From (9) and (10) we have

$$\frac{F(z+\Delta z) - F(z)}{\Delta z} - f(z) = \frac{1}{\Delta z}\int_{z}^{z+\Delta z} [f(s) - f(z)]\,ds.$$

Now f is continuous at the point z. This means for any $\varepsilon > 0$ there exists a $\delta > 0$ so that $|f(s) - f(z)| < \varepsilon$ whenever $|s - z| < \delta$. Consequently, if we choose Δz so that $|\,\Delta z\,| < \delta$, it follows from the *ML*-inequality of Section 5.2 that

$$\left|\frac{F(z+\Delta z) - F(z)}{\Delta z} - f(z)\right| = \left|\frac{1}{\Delta z}\int_{z}^{z+\Delta z} [f(s) - f(z)]\,ds\right|$$

$$= \left|\frac{1}{\Delta z}\right|\left|\int_{z}^{z+\Delta z} [f(s) - f(z)]\,ds\right| \le \left|\frac{1}{\Delta z}\right| \varepsilon\,|\Delta z| = \varepsilon.$$

Hence, we have shown that

$$\lim_{\Delta z \to 0} \frac{F(z+\Delta z) - F(z)}{\Delta z} = f(z) \quad \text{or} \quad F'(z) = f(z).$$

☰

If f is an analytic function in a simply connected domain D, it is necessarily continuous throughout D. This fact, when put together with the results in Theorem 5.4.1 and (7), leads to a theorem which states that an analytic function possesses an analytic antiderivative.

THEOREM 5.4.3 Existence of an Antiderivative

Suppose that a function f is analytic in a simply connected domain D. Then f has an antiderivative in D; that is, there exists a function F such that $F'(z) = f(z)$ for all z in D.

Be careful when using Ln z as an antiderivative of $1/z$. ▶

In (21) of Section 4.1 we saw for $|z| > 0$, $-\pi < \arg(z) < \pi$, that $1/z$ is the derivative of Ln z. This means that under some circumstances Ln z is an antiderivative of $1/z$. But care must be exercised in using this result. For example, suppose D is the entire complex plane without the origin. The function $1/z$ is analytic in this *multiply connected* domain. If C is any simple closed contour containing the origin, it does

not follow from (5) that $\oint_C dz/z = 0$. In fact, from (6) of Section 5.3 with $n = 1$ and the identification of $z_0 = 0$, we see that

$$\oint_C \frac{1}{z}\,dz = 2\pi i.$$

In this case, Ln z is not an antiderivative of $1/z$ in D since Ln z is not analytic in D. Recall, Ln z fails to be analytic on the nonpositive real axis which is the branch cut of the principal branch $f_1(z)$ of the logarithm. See page 160 in Section 4.1.

EXAMPLE 4 Using the Logarithmic Function

Evaluate $\displaystyle\int_C \frac{1}{z}\,dz$, where C is the contour shown in FIGURE 5.4.4.

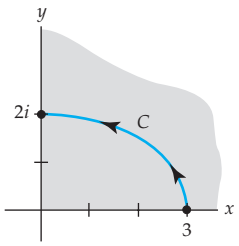


FIGURE 5.4.4 Contour for Example 4

Solution Suppose that D is the simply connected domain defined by $x > 0$, $y > 0$, in other words, D is the first quadrant in the z-plane. In this case, Ln z *is* an antiderivative of $1/z$ since both these functions are analytic in D. Hence by (4),

$$\int_3^{2i} \frac{1}{z}\,dz = \text{Ln } z\Big|_3^{2i} = \text{Ln } 2i - \text{Ln } 3.$$

From (14) of Section 4.1,

$$\text{Ln } 2i = \log_e 2 + \frac{\pi}{2}i \qquad \text{and} \qquad \text{Ln } 3 = \log_e 3$$

$\log_e 2 - \log_e 3 = \log_e \frac{2}{3}$ ▶ and so, $$\int_3^{2i} \frac{1}{z}\,dz = \log_e \frac{2}{3} + \frac{\pi}{2}i \approx -0.4055 + 1.5708i.$$ ☰

EXAMPLE 5 Using an Antiderivative of $z^{-1/2}$

Evaluate $\displaystyle\int_C \frac{1}{z^{1/2}}\,dz$, where C is the line segment between $z_0 = i$ and $z_1 = 9$.

Solution Throughout, we take $f_1(z) = z^{1/2}$ to be the principal branch of the square root function. In the domain $|z| > 0$, $-\pi < \arg(z) < \pi$, the function $f_1(z) = 1/z^{1/2} = z^{-1/2}$ is analytic and possesses the antiderivative $F(z) = 2z^{1/2}$ (see (8) in Section 4.2). Hence,

$$\int_i^9 \frac{1}{z^{1/2}}dz = 2z^{1/2}\Big|_i^9 = 2\left[3 - \overbrace{\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right)}^{\text{See Problem 5 in Exercises 1.4}}\right]$$

$$= (6 - \sqrt{2}) - i\sqrt{2}.$$ ☰

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) In the study of techniques of integration in calculus you learned that indefinite integrals of certain kinds of products could be evaluated by integration by parts:

$$\int f(x)g'(x)dx = f(x)g(x) - \int g(x)f'(x)dx. \tag{11}$$

You undoubtedly have used (11) in the more compact form $\int u\,dv = uv - \int v\,du$. Formula (11) carries over to complex analysis. Suppose f and g are analytic in a simply connected domain D. Then

$$\int f(z)g'(z)dz = f(z)g(z) - \int g(z)f'(z)dz. \tag{12}$$

In addition, if z_0 and z_1 are the initial and terminal points of a contour C lying entirely in D, then

$$\int_{z_0}^{z_1} f(z)g'(z)\,dz = f(z)g(z)\Big|_{z_0}^{z_1} - \int_{z_0}^{z_1} g(z)f'(z)\,dz. \tag{13}$$

These results can be proved in a straightforward manner using Theorem 5.4.2 on the function $\dfrac{d}{dz}fg$. See Problems 21–24 in Exercises 5.4.

(*ii*) If f is a real function continuous on the closed interval $[a, b]$, then there exists a number c in the open interval (a, b) such that

$$\int_a^b f(x)\,dx = f(c)(b - a). \tag{14}$$

The result in (14) is known as the **mean-value theorem** for definite integrals. If f is a complex function analytic in a simply connected domain D, it is continuous at every point on a contour C in D with initial point z_0 and terminal point z_1. One might expect a result parallel to (14) for an integral $\int_{z_0}^{z_1} f(z)\,dz$. However, there is no such complex counterpart.

EXERCISES 5.4

Answers to selected odd-numbered problems begin on page ANS-15.

In Problems 1 and 2, evaluate the given integral, where the contour C is given in the figure, (**a**) by using an alternative path of integration and (**b**) by using Theorem 5.4.2.

1. $\displaystyle\int_C (4z - 1)\,dz$

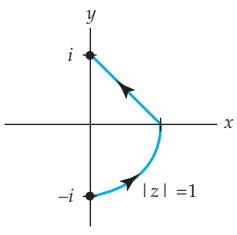


FIGURE 5.4.5 Figure for Problem 1

2. $\displaystyle\int_C e^z\,dz$

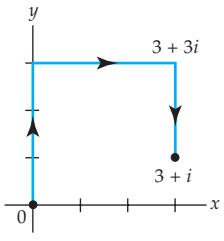


FIGURE 5.4.6 Figure for Problem 2

In Problems 3 and 4, evaluate the given integral along the indicated contour C.

3. $\displaystyle\int_C 2z\,dz$, where C is $z(t) = 2t^3 + i(t^4 - 4t^3 + 2)$, $-1 \le t \le 1$

4. $\displaystyle\int_C 2z\,dz$, where C is $z(t) = 2\cos^3 \pi t - i \sin^2 \frac{\pi}{4}t$, $0 \le t \le 2$

In Problems 5–20, use Theorem 5.4.2 to evaluate the given integral. Write each answer in the form $a + ib$.

5. $\displaystyle\int_0^{3+i} z^2\,dz$

6. $\displaystyle\int_{-2i}^{1} (3z^2 - 4z + 5i)\,dz$

7. $\displaystyle\int_{1-i}^{1+i} z^3\,dz$

8. $\displaystyle\int_{-3i}^{2i} (z^3 - z)\,dz$

9. $\displaystyle\int_{-i/2}^{1-i} (2z + 1)^2\,dz$

10. $\displaystyle\int_{1}^{i} (iz + 1)^3\,dz$

11. $\displaystyle\int_{i/2}^{i} e^{\pi z}\,dz$

12. $\displaystyle\int_{1-i}^{1+2i} ze^{z^2}\,dz$

13. $\displaystyle\int_{\pi}^{\pi+2i} \sin\frac{z}{2}\,dz$

14. $\displaystyle\int_{1-2i}^{\pi i} \cos z\,dz$

15. $\displaystyle\int_{\pi i}^{2\pi i} \cosh z\,dz$

16. $\displaystyle\int_{i}^{1+(\pi/2)i} \sinh 3z\,dz$

17. $\displaystyle\int_C \frac{1}{z}\,dz$, C is the arc of the circle $z = 4e^{it}$, $-\pi/2 \le t \le \pi/2$

18. $\displaystyle\int_C \frac{1}{z}\,dz$, C is the line segment between $1 + i$ and $4 + 4i$

19. $\displaystyle\int_{-4i}^{4i} \frac{1}{z^2}\,dz$, C is any contour not passing through the origin

20. $\displaystyle\int_{1-i}^{1+\sqrt{3}i} \left(\frac{1}{z} + \frac{1}{z^2}\right) dz$, C is any contour in the right half-plane $\text{Re}(z) > 0$

In Problems 21–24, use integration by parts (13) to evaluate the given integral. Write each answer in the form $a + ib$.

21. $\displaystyle\int_{\pi}^{i} e^z \cos z\,dz$

22. $\displaystyle\int_{0}^{i} z \sin z\,dz$

23. $\displaystyle\int_{i}^{1+i} ze^z\,dz$

24. $\displaystyle\int_{0}^{\pi i} z^2 e^z\,dz$

In Problems 25 and 26, use Theorem 5.4.2 to evaluate the given integral. In each integral $z^{1/2}$ is the principal branch of the square root function. Write each answer in the form $a + ib$.

25. $\displaystyle\int_C \frac{1}{4z^{1/2}}\,dz$, C is the arc of the circle $z = 4e^{it}$, $-\pi/2 \le t \le \pi/2$

26. $\displaystyle\int_C 3z^{1/2}\,dz$, C is the line segment between $z_0 = 1$ and $z_1 = 9i$

Focus on Concepts

27. Find an antiderivative of $f(z) = \sin z^2$. Do not think profound thoughts.

28. Give a domain D over which $f(z) = z(z + 1)^{1/2}$ is analytic. Then find an antiderivative of f in D.

29. Let $\alpha = a + ib$ be a complex constant.

(**a**) Use Theorem 5.4.2 to evaluate $\displaystyle\int_{x_0}^{x} e^{\alpha z}\,dz$ where x_0 and x are real values.

(**b**) Explain how part (a) and Theorem 5.2.1 with the parametrization $z(t) = t, x_0 \le t \le x$ can be used to derive the (real) integral formula

$$\int e^{ax} \cos bx\,dx = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2 + b^2} + C.$$

30. Find $\int e^{ax} \sin bx\, dx$. [*Hint*: See Problem 29.]

31. In this problem you are guided through the proof that if $\phi(x, y)$ is a harmonic function in a simply connected domain D, then ϕ has a harmonic conjugate ψ in D.

Proof Assume that $\phi(x, y)$ is a harmonic function in the simply connected domain D. Since ϕ is harmonic, it has continuous first and second order partial derivatives in D. Now define a complex function in D by

$$g(z) = \frac{\partial \phi}{\partial x}(x, y) - i\frac{\partial \phi}{\partial y}(x, y).$$

(**a**) Use Theorem 3.3.2 to show that $g(z)$ is analytic in D.

(**b**) Why must $g(z)$ now have an antiderivative $G(z)$ in D?

(**c**) Complete the proof by showing that $\psi(x, y) = \text{Im}(G(z))$ is a harmonic conjugate of $\phi(x, y)$ in D.

32. Suppose $\phi(x, y)$ is a harmonic function in a simply connected domain D containing the point $z_0 = x_0 + iy_0$. Use Problem 31, (8) from Section 5.4, and (9) from Section 5.2 to show that its harmonic conjugate is given by the line integral

$$\psi(x, y) = \int_{(x_0, y_0)}^{(x, y)} -\frac{\partial \phi}{\partial t}\, ds + \frac{\partial \phi}{\partial s}\, dt.$$

5.5 Cauchy's Integral Formulas and Their Consequences

INTRODUCTION In the last two sections we saw the importance of the Cauchy-Goursat theorem in the evaluation of contour integrals. In this section we are going to examine several more consequences of the Cauchy-Goursat theorem. Unquestionably, the most significant of these is the following result:

> *The value of a analytic function f at any point z_0 in a simply connected domain can be represented by a contour integral.*

After establishing this proposition we shall use it to further show that:

> *An analytic function f in a simply connected domain possesses derivatives of all orders.*

The ramifications of these two results alone will keep us busy not only for the remainder of this section but in the next chapter as well.

5.5.1 Cauchy's Two Integral Formulas

First Formula If f is analytic in a simply connected domain D and z_0 is any point in D, the quotient $f(z)/(z - z_0)$ is not defined at z_0 and hence is *not* analytic in D. Therefore, we *cannot* conclude that the integral of $f(z)/(z - z_0)$ around a simple closed contour C that contains z_0 is zero by the Cauchy-Goursat theorem. Indeed, as we shall now see, the integral of $f(z)/(z - z_0)$ around C has the value $2\pi i f(z_0)$. The first of two remarkable formulas is known simply as the **Cauchy integral formula**.

THEOREM 5.5.1 Cauchy's Integral Formula

Suppose that f is analytic in a simply connected domain D and C is any simple closed contour lying entirely within D. Then for any point z_0 within C,

$$f(z_0) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z - z_0}\, dz. \tag{1}$$

Proof Let D be a simply connected domain, C a simple closed contour in D, and z_0 an interior point of C. In addition, let C_1 be a circle centered at z_0 with radius small enough so that C_1 lies within the interior of C. By the principle of deformation of contours, (5) of Section 5.3, we can write

$$\oint_C \frac{f(z)}{z - z_0}\, dz = \oint_{C_1} \frac{f(z)}{z - z_0}\, dz. \tag{2}$$

We wish to show that the value of the integral on the right is $2\pi i f(z_0)$. To this end we add and subtract the constant $f(z_0)$ in the numerator of the integrand,

$$\begin{aligned}\oint_{C_1} \frac{f(z)}{z - z_0}\, dz &= \oint_{C_1} \frac{f(z_0) - f(z_0) + f(z)}{z - z_0}\, dz. \\ &= f(z_0) \oint_{C_1} \frac{1}{z - z_0}\, dz + \oint_{C_1} \frac{f(z) - f(z_0)}{z - z_0}\, dz. \end{aligned} \tag{3}$$

From (6) of Section 5.3 we know that

$$\oint_{C_1} \frac{1}{z - z_0}\, dz = 2\pi i \tag{4}$$

and so (3) becomes

$$\oint_{C_1} \frac{f(z)}{z - z_0}\, dz = 2\pi i\, f(z_0) + \oint_{C_1} \frac{f(z) - f(z_0)}{z - z_0}\, dz. \tag{5}$$

Since f is continuous at z_0, we know that for any arbitrarily small $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - f(z_0)| < \varepsilon$ whenever $|z - z_0| < \delta$. In particular, if we choose the circle C_1 to be $|z - z_0| = \frac{1}{2}\delta < \delta$, then by the *ML*-inequality (Theorem 5.2.3) the absolute value of the integral on the right side of the equality in (5) satisfies

$$\left| \oint_{C_1} \frac{f(z) - f(z_0)}{z - z_0}\, dz \right| \le \frac{\varepsilon}{\delta/2} 2\pi \left(\frac{\delta}{2}\right) = 2\pi\varepsilon.$$

In other words, the absolute value of the integral can be made arbitrarily small by taking the radius of the circle C_1 to be sufficiently small. This can happen only if the integral is 0. Thus (5) is $\oint_{C_1} \frac{f(z)}{z - z_0}\, dz = 2\pi i\, f(z_0)$. The theorem is proved by dividing both sides of the last result by $2\pi i$. ☰

Note ▶ Because the symbol z represents a point on the contour C, (1) indicates that *the values of an analytic function f at points z_0 inside a simple closed contour C are determined by the values of f on the contour C.*

Cauchy's integral formula (1) can be used to evaluate contour integrals. Since we often work problems without a simply connected domain explicitly defined, a more practical restatement of Theorem 5.5.1 is:

- *If f is analytic at all points within and on a simple closed contour C, and z_0 is any point interior to C, then* $f(z_0) = \dfrac{1}{2\pi i} \oint_C \dfrac{f(z)}{z - z_0}\, dz.$

EXAMPLE 1 Using Cauchy's Integral Formula

Evaluate $\oint_C \dfrac{z^2 - 4z + 4}{z + i}\, dz$, where C is the circle $|z| = 2$.

Solution First, we identify $f(z) = z^2 - 4z + 4$ and $z_0 = -i$ as a point within the circle C. Next, we observe that f is analytic at all points within and on the contour C.

Thus, by the Cauchy integral formula (1) we obtain

$$\oint_C \frac{z^2 - 4z + 4}{z + i}\, dz = 2\pi i f(-i) = 2\pi i(3 + 4i) = \pi(-8 + 6i).$$

☰

EXAMPLE 2 **Using Cauchy's Integral Formula**

Evaluate $\oint_C \frac{z}{z^2 + 9}\, dz$, where C is the circle $|z - 2i| = 4$.

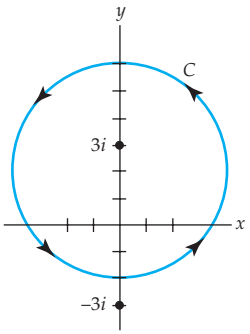


FIGURE 5.5.1 Contour for Example 2

Solution By factoring the denominator as $z^2 + 9 = (z - 3i)(z + 3i)$ we see that $3i$ is the only point within the closed contour C at which the integrand fails to be analytic. See FIGURE 5.5.1. Then by rewriting the integrand as

$$\frac{z}{z^2 + 9} = \frac{\left\{\dfrac{z}{z + 3i}\right\} f(z)}{z - 3i},$$

we can identify $f(z) = z/(z + 3i)$. The function f is analytic at all points within and on the contour C. Hence, from Cauchy's integral formula (1) we have

$$\oint_C \frac{z}{z^2 + 9}\, dz = \int_C \frac{\dfrac{z}{z + 3i}}{z - 3i}\, dz = 2\pi i\, f(3i) = 2\pi i \frac{3i}{6i} = \pi i.$$

☰

Second Formula We shall now build on Theorem 5.5.1 by using it to prove that the values of the derivatives $f^{(n)}(z_0), n = 1, 2, 3, \ldots$ of an analytic function are also given by a integral formula. This second integral formula is similar to (1) and is known by the name **Cauchy's integral formula for derivatives**.

THEOREM 5.5.2 **Cauchy's Integral Formula for Derivatives**

Suppose that f is analytic in a simply connected domain D and C is any simple closed contour lying entirely within D. Then for any point z_0 within C,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}}\, dz. \tag{6}$$

Partial Proof We will prove (6) only for the case $n = 1$. The remainder of the proof can be completed using the principle of mathematical induction. We begin with the definition of the derivative and (1):

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{1}{2\pi i\, \Delta z} \left[\oint_C \frac{f(z)}{z - (z_0 + \Delta z)}\, dz - \oint_C \frac{f(z)}{z - z_0}\, dz \right]$$

$$= \lim_{\Delta z \to 0} \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0 - \Delta z)(z - z_0)}\, dz.$$

Before continuing, let us set out some preliminaries. Continuity of f on the contour C guarantees that f is bounded (see page 114 of Section 3.1), that is, there exists a

real number M such that $|f(z)| \le M$ for all points z on C. In addition, let L be the length of C and let δ denote the shortest distance between points on C and the point z_0. Thus for all points z on C we have

$$|z - z_0| \ge \delta \qquad \text{or} \qquad \frac{1}{|z - z_0|^2} \le \frac{1}{\delta^2}.$$

Furthermore, if we choose $|\Delta z| \le \frac{1}{2}\delta$, then by (10) of Section 1.2,

$$|z - z_0 - \Delta z| \ge \big|\, |z - z_0| - |\Delta z| \,\big| \ge \delta - |\Delta z| \ge \tfrac{1}{2}\delta$$

and so,

$$\frac{1}{|z - z_0 - \Delta z|} \le \frac{2}{\delta}.$$

Now,

$$\left| \oint_C \frac{f(z)}{(z - z_0)^2}\, dz - \oint_C \frac{f(z)}{(z - z_0 - \Delta z)(z - z_0)}\, dz \right| = \left| \oint_C \frac{-\Delta z\, f(z)}{(z - z_0 - \Delta z)(z - z_0)^2}\, dz \right| \le \frac{2ML\,|\Delta z|}{\delta^3}.$$

Because the last expression approaches zero as $\Delta z \to 0$, we have shown that

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2}\, dz,$$

which is (6) for $n = 1$. ☰

Like (1), formula (6) can be used to evaluate integrals.

EXAMPLE 3 Using Cauchy's Integral Formula for Derivatives

Evaluate $\displaystyle\oint_C \frac{z + 1}{z^4 + 2iz^3}\, dz$, where C is the circle $|z| = 1$.

Solution Inspection of the integrand shows that it is not analytic at $z = 0$ and $z = -2i$, but only $z = 0$ lies within the closed contour. By writing the integrand as

$$\frac{z + 1}{z^4 + 2iz^3} = \frac{\dfrac{z + 1}{z + 2i}}{z^3}$$

we can identify, $z_0 = 0$, $n = 2$, and $f(z) = (z + 1)/(z + 2i)$. The quotient rule gives $f''(z) = (2 - 4i)/(z + 2i)^3$ and so $f''(0) = (2i - 1)/4i$. Hence from (6) we find

$$\oint_C \frac{z + 1}{z^4 + 4z^3}\, dz = \frac{2\pi i}{2!} f''(0) = -\frac{\pi}{4} + \frac{\pi}{2} i.$$ ☰

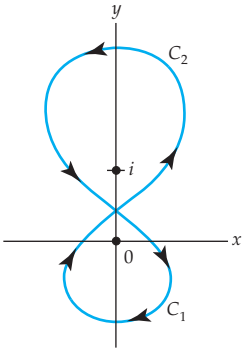


FIGURE 5.5.2 Contour for Example 4

EXAMPLE 4 Using Cauchy's Integral Formula for Derivatives

Evaluate $\displaystyle\int_C \frac{z^3 + 3}{z(z - i)^2}\, dz$, where C is the figure-eight contour shown in FIGURE 5.5.2.

Solution Although C is not a simple closed contour, we can think of it as the union of two simple closed contours C_1 and C_2 as indicated in Figure 5.5.2. Since the arrows

on C_1 flow clockwise or in the negative direction, the opposite curve $-C_1$ has positive orientation. Hence, we write

$$\int_C \frac{z^3+3}{z(z-i)^2}\,dz = \int_{C_1} \frac{z^3+3}{z(z-i)^2}\,dz + \int_{C_2} \frac{z^3+3}{z(z-i)^2}\,dz$$

$$= -\oint_{-C_1} \frac{\dfrac{z^3+3}{(z-i)^2}}{z}\,dz + \oint_{C_2} \frac{\dfrac{z^3+3}{z}}{(z-i)^2}\,dz = -I_1 + I_2,$$

and we are in a position to use both formulas (1) and (6).

To evaluate I_1 we identify $z_0 = 0$, $f(z) = (z^3+3)/(z-i)^2$, and $f(0) = -3$. By (1) it follows that

$$I_1 = \oint_{-C_1} \frac{\dfrac{z^3+3}{(z-i)^2}}{z}\,dz = 2\pi i\, f(0) = 2\pi i(-3) = -6\pi i.$$

To evaluate I_2 we now identify $z_0 = i$, $n = 1$, $f(z) = (z^3+3)/z$, $f'(z) = (2z^3-3)/z^2$, and $f'(i) = 3+2i$. From (6) we obtain

$$I_2 = \oint_{C_2} \frac{\dfrac{z^3+3}{z}}{(z-i)^2}\,dz = \frac{2\pi i}{1!}\, f'(i) = 2\pi i(3+2i) = -4\pi + 6\pi i.$$

Finally, we get

$$\int_C \frac{z^3+3}{z(z-i)^2}\,dz = -I_1 + I_2 = 6\pi i + (-4\pi + 6\pi i) = -4\pi + 12\pi i.$$

5.5.2 Some Consequences of the Integral Formulas

An immediate and important corollary to Theorem 5.5.2 is summarized next.

THEOREM 5.5.3 Derivative of an Analytic Function Is Analytic

Suppose that f is analytic in a simply connected domain D. Then f possesses derivatives of all orders at every point z in D. The derivatives f', f'', f''', ... are analytic functions in D.

If a function $f(z) = u(x, y) + iv(x, y)$ is analytic in a simply connected domain D, we have just seen its derivatives of all orders exist at any point z in D and so f', f'', f''', ... are continuous. From

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$$

$$f''(z) = \frac{\partial^2 u}{\partial x^2} + i\frac{\partial^2 v}{\partial x^2} = \frac{\partial^2 v}{\partial y \partial x} - i\frac{\partial^2 u}{\partial y \partial x}$$

$$\vdots$$

we can also conclude that the real functions u and v have continuous partial derivatives of all orders at a point of analyticity.

Cauchy's Inequality We next present an inequality derived from the Cauchy integral formula for derivatives.

THEOREM 5.5.4 Cauchy's Inequality

Suppose that f is analytic in a simply connected domain D and C is a circle defined by $|z - z_0| = r$ that lies entirely in D. If $|f(z)| \leq M$ for all points z on C, then

$$\left|f^{(n)}(z_0)\right| \leq \frac{n!M}{r^n}. \tag{7}$$

Proof From the hypothesis,

$$\left|\frac{f(z)}{(z - z_0)^{n+1}}\right| = \frac{|f(z)|}{r^{n+1}} \leq \frac{M}{r^{n+1}}.$$

Thus, from (6) and the *ML*-inequality (Theorem 5.2.3), we have

$$\left|\, f^{(n)}(z_0)\,\right| = \left|\frac{n!}{2\pi i}\oint_C \frac{f(z)}{(z - z_0)^{n+1}}dz\right| \leq \frac{n!}{2\pi}\frac{M}{r^{n+1}}2\pi r = \frac{n!M}{r^n}.$$

The number M in Theorem 5.5.4 depends on the circle $|z - z_0| = r$. But notice in (7) that if $n = 0$, then $M \geq |\, f(z_0)\,|$ for *any* circle C centered at z_0 as long as C lies within D. In other words, an upper bound M of $|f(z)|$ on C cannot be smaller than $|\, f(z_0)\,|$.

Liouville's Theorem Cauchy's inequality (7) is a key ingredient in the proof of the next result. Although it bears the name "Liouville's theorem," it probably was first proved by Cauchy. The gist of the theorem is that an entire function f, one that is analytic for all z, cannot be bounded unless f itself is a constant.

THEOREM 5.5.5 Liouville's Theorem

The only bounded entire functions are constants.

Proof Suppose f is an entire function and is bounded, that is, $|f(z)| \leq M$ for all z. Then for any point z_0, (7) gives $\left|\, f'(z_0)\,\right| \leq M/r$. By making r arbitrarily large we can make $\left|\, f'(z_0)\,\right|$ as small as we wish. This means $f'(z_0) = 0$ for all points z_0 in the complex plane. Hence, by Theorem 3.3.3(*ii*), f must be a constant.

Fundamental Theorem of Algebra Theorem 5.5.5 enables us to establish a result usually learned—but never proved—in elementary algebra.

THEOREM 5.5.6 Fundamental Theorem of Algebra

If $p(z)$ is a nonconstant polynomial, then the equation $p(z) = 0$ has at least one root.

Proof Let us suppose that the polynomial $p(z) = a_n z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0$, $n > 0$, is not 0 for any complex number z. This implies that the reciprocal of p,

$f(z) = 1/p(z)$, is an entire function. Now

$$|f(z)| = \frac{1}{\left|a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0\right|}$$

$$= \frac{1}{|z|^n \left|a_n + \left(a_{n-1}/z + \cdots + a_1/z^{n-1} + a_0/z^n\right)\right|}.$$

Choose a real number $M > 1$ such that $M > |2na_j/a_n|$ for $j = 0, 1, \ldots, n-1$. Then for $|z| > M$ we have $|a_j/z^{n-j}| < |a_n|/2n$, and so

$$\left|a_{n-1}/z + \cdots + a_1/z^{n-1} + a_0/z\right| \leq |a_n|/2n + \cdots + |a_n|/2n = |a_n|/2.$$

From inequality (8) of Section 1.2 we then have:

$$|f(z)| \leq \frac{1}{|z|^n \left||a_n| - \left|a_{n-1}/z + \cdots + a_1/z^{n-1} + a_0/z^n\right|\right|}$$

$$\leq \frac{1}{M^n \left||a_n| - |a_n|/2\right|} = \frac{2}{M^n |a_n|}.$$

Thus, we see that f is bounded on the exterior of the disk $|z| \leq M$. However, since f is continuous it must also be bounded on the disk $|z| \leq M$ (see Section 3.1). We conclude that the function f is a bounded entire function. It then follows from Liouville's theorem that f is a constant, and therefore p is a constant. But this is a contradiction to our underlying assumption that p was *not* a constant polynomial. We conclude that there must exist at least one number z for which $p(z) = 0$. ☰

It is left as an exercise to show, using Theorem 5.5.6, that if $p(z)$ is a nonconstant polynomial of degree n, then $p(z) = 0$ has exactly n roots (counting multiple roots). See Problem 29 in Exercises 5.5.

Morera's Theorem The proof of the next theorem enshrined the name of the Italian mathematician Giacinto Morera forever in texts on complex analysis. Morera's theorem, which gives a sufficient condition for analyticity, is often taken to be the converse of the Cauchy-Goursat theorem. For its proof we return to Theorem 5.5.3.

THEOREM 5.5.7 Morera's Theorem

If f is continuous in a simply connected domain D and if $\oint_C f(z)\,dz = 0$ for every closed contour C in D, then f is analytic in D.

Proof By the hypotheses of continuity of f and $\oint_C f(z)\,dz = 0$ for every closed contour C in D, we conclude that $\int_C f(z)\,dz$ is independent of the path. In the proof of (7) of Section 5.4 we then saw that the function F defined by $F(z) = \int_{z_0}^{z} f(s)\,ds$ (where s denotes a complex variable, z_0 is a fixed point in D, and z represents any point in D) is an antiderivative of f; that is, $F'(z) = f(z)$. Hence, F is analytic in D. In addition, $F'(z)$ is analytic in view of Theorem 5.5.3. Since $f(z) = F'(z)$, we see that f is analytic in D. ☰

An alternative proof of this last result is outlined in Problem 31 in Exercises 5.5.

We could go on at length stating more and more results whose proofs rest on a foundation of theory that includes the Cauchy-Goursat theorem and the Cauchy integral formulas. But we shall stop after one more theorem.

In Section 3.1 we saw that if a function f is continuous on a closed and bounded region R, then f is bounded; that is, there is some constant M such that $|f(z)| \leq M$ for z in R. If the boundary of R is a simple closed curve C, then the next theorem, which we present without proof, tells us that $|f(z)|$ assumes its maximum value at some point z on the boundary C.

THEOREM 5.5.8 Maximum Modulus Theorem

Suppose that f is analytic and nonconstant on a closed region R bounded by a simple closed curve C. Then the modulus $|f(z)|$ attains its maximum on C.

If the stipulation that $f(z) \neq 0$ for all z in R is added to the hypotheses of Theorem 5.5.8, then the modulus $|f(z)|$ also attains its *minimum* on C. See Problems 27 and 33 in Exercises 5.5.

EXAMPLE 5 Maximum Modulus

Find the maximum modulus of $f(z) = 2z + 5i$ on the closed circular region defined by $|z| \leq 2$.

Solution From (2) of Section 1.2 we know that $|z|^2 = z\bar{z}$. By replacing the symbol z by $2z + 5i$ we have

$$|2z + 5i|^2 = (2z + 5i)\overline{(2z + 5i)} = (2z + 5i)(2\bar{z} - 5i) = 4z\bar{z} - 10i(z - \bar{z}) + 25. \tag{8}$$

But from (6) of Section 1.1, $\bar{z} - z = 2i\,\mathrm{Im}(z)$, and so (8) is

$$|2z + 5i|^2 = 4|z|^2 + 20\,\mathrm{Im}(z) + 25. \tag{9}$$

Because f is a polynomial, it is analytic on the region defined by $|z| \leq 2$. By Theorem 5.5.8, $\max_{|z|\leq 2} |\,2z + 5i\,|$ occurs on the boundary $|z| = 2$. Therefore, if $|z| = 2$, then (9) yields

$$|\,2z + 5i\,| = \sqrt{41 + 20\,\mathrm{Im}(z)}. \tag{10}$$

The last expression attains its maximum when $\mathrm{Im}(z)$ attains its maximum on $|z| = 2$, namely, at the point $z = 2i$. Thus, $\max_{|z|\leq 2} |\,2z + 5i\,| = \sqrt{81} = 9$. ☰

Note in Example 5 that $f(z) = 0$ only at $z = -\frac{5}{2}i$ and that this point is outside the region defined by $|z| \leq 2$. Hence we can conclude that (10) attains its minimum when $\mathrm{Im}(z)$ attains its minimum on $|z| = 2$ at $z = -2i$. As a result, $\min_{|z|\leq 2} |\,2z + 5i\,| = \sqrt{1} = 1$.

Remarks **COMPARISON WITH REAL ANALYSIS**

(*i*) As a consequence of Theorem 5.5.3, if f is differentiable at all points in some domain, then the derivatives f', f'', f''', ... all exist on the domain. In real analysis this need not be true. For example, the function $f(x) = x^{5/3}$ is differentiable on the entire real line. However, $f''(x) = \frac{10}{9}x^{-1/3}$ does not exist at $x = 0$.

(*ii*) In real analysis there are many infinitely differentiable functions that are bounded on a closed interval $[a, b]$, that is, $|f(x)| \leq M$ for all $a \leq x \leq b$. For example, $\sin x$ is infinitely differentiable and bounded on $[-\pi, \pi]$ since $|\sin x| \leq 1$ for all x. The absolute value $|\sin x|$, however, does not attain its maximum on the boundary of the closed interval since $|\sin(-\pi)| = |\sin \pi| = 0$. Compare this fact with the Maximum Modulus Theorem 5.5.8.

EXERCISES 5.5

Answers to selected odd-numbered problems begin on page ANS-15.

5.5.1 Cauchy's Integral Formulas

In Problems 1–22, use Theorems 5.5.1 and 5.5.2, when appropriate, to evaluate the given integral along the indicated closed contour(s).

1. $\displaystyle\oint_C \frac{4}{z - 3i}\, dz;\ |z| = 5$

2. $\displaystyle\oint_C \frac{z^2}{(z - 3i)^2}\, dz;\ |z| = 5$

3. $\displaystyle\oint_C \frac{e^z}{z - \pi i}\, dz;\ |z| = 4$

4. $\displaystyle\oint_C \frac{1 + e^z}{z}\, dz;\ |z| = 1$

5. $\displaystyle\oint_C \frac{z^2 - 3z + 4i}{z + 2i}\, dz;\ |z| = 3$

6. $\displaystyle\oint_C \frac{\cos z}{3z - \pi}\, dz;\ |z| = 1.1$

7. $\displaystyle\oint_C \frac{z^2}{z^2 + 4}\, dz;$ **(a)** $|z - i| = 2,$ **(b)** $|z + 2i| = 1$

8. $\displaystyle\oint_C \frac{z^2 + 3z + 2i}{z^2 + 3z - 4}\, dz;$ **(a)** $|z| = 2,$ **(b)** $|z + 5| = \frac{3}{2}$

9. $\displaystyle\oint_C \frac{z^2 + 4}{z^2 - 5iz - 4}\, dz;\ |z - 3i| = 1.3$

10. $\displaystyle\oint_C \frac{\sin z}{z^2 + \pi^2}\, dz;\ |z - 2i| = 2$

11. $\displaystyle\oint_C \frac{e^{z^2}}{(z - i)^3}\, dz;\ |z - i| = 1$

12. $\displaystyle\oint_C \frac{z}{(z + i)^4}\, dz;\ |z| = 2$

13. $\displaystyle\oint_C \frac{\cos 2z}{z^5}\, dz;\ |z| = 1$

14. $\displaystyle\oint_C \frac{e^{-z} \sin z}{z^3}\, dz;\ |z - 1| = 3$

15. $\displaystyle\oint_C \frac{2z + 5}{z^2 - 2z}\, dz;$ **(a)** $|z| = \frac{1}{2},$ **(b)** $|z + 1| = 2$ **(c)** $|z - 3| = 2,$ **(d)** $|z + 2i| = 1$

16. $\displaystyle\oint_C \frac{z}{(z - 1)(z - 2)}\, dz;$ **(a)** $|z| = \frac{1}{2},$ **(b)** $|z + 1| = 1$ **(c)** $|z - 1| = \frac{1}{2},$ **(d)** $|z| = 4$

17. $\displaystyle\oint_C \frac{z + 2}{z^2(z - 1 - i)}\, dz;$ **(a)** $|z| = 1,$ **(b)** $|z - 1 - i| = 1$

18. $\displaystyle\oint_C \frac{1}{z^3(z - 4)}\, dz;$ **(a)** $|z| = 1,$ **(b)** $|z - 2| = 1$

19. $\displaystyle\oint_C \left(\frac{e^{2iz}}{z^4} - \frac{z^4}{(z - i)^3}\right) dz;\ |z| = 6$

20. $\displaystyle\oint_C \left(\frac{\cosh z}{(z-\pi)^3} - \frac{\sin^2 z}{(2z-\pi)^3} \right) dz;\ |z| = 3$

21. $\displaystyle\oint_C \frac{1}{z^3(z-1)^2}\, dz;\ |z-2| = 5$

22. $\displaystyle\oint_C \frac{1}{z^2(z^2+1)}\, dz;\ |z-i| = \tfrac{3}{2}$

In Problems 23 and 24, evaluate the given integral, where C is the figure-eight contour in the figure.

23. $\displaystyle\oint_C \frac{3z+1}{z(z-2)^2}\, dz$

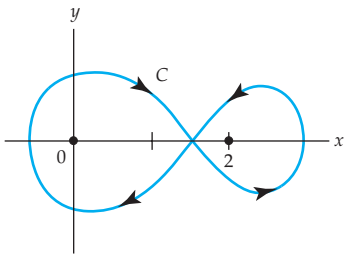


FIGURE 5.5.3 Figure for Problem 23

24. $\displaystyle\oint_C \frac{e^{iz}}{(z^2+1)^2}\, dz$

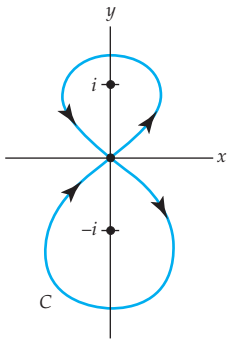


FIGURE 5.5.4 Figure for Problem 24

5.5.2 Some Consequences of the Integral Formulas

In Problems 25 and 26, proceed as in Example 5 to find the maximum modulus of the given function on indicated closed circular region.

25. $f(z) = -iz + i;\ |z| \le 5$

26. $f(z) = z^2 + 4z;\ |z| \le 1$

27. Suppose the boundary C of the closed circular region R defined by $|z| \le 1$ is parametrized by $x = \cos t$, $y = \sin t$, $0 \le t \le 2\pi$. By considering $|f(z(t))|$, find the maximum modulus and the minimum modulus of the given analytic function f and the points z on C that give these values.

(a) $f(z) = (iz+3)^2$

(b) $f(z) = (z - 2 - 2\sqrt{3}\,i)^2$

(c) $f(z) = -2iz^2 + 5$

[*Hint*: In parts (b) and (c), it may help to recall from calculus how to find the relative extrema of a real-valued function of real variable t.]

Focus on Concepts

28. (***Cauchy's Integral Formula***) Suppose f is analytic within and on a circle C of radius r with center at z_0. Use (1) to obtain

$$f(z_0) = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + re^{i\theta})\, d\theta.$$

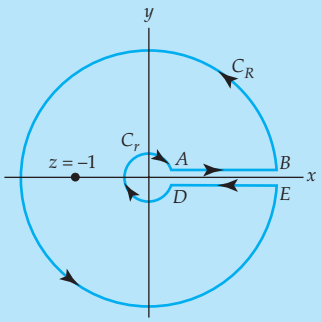
This result is known as **Gauss' mean-value theorem** and shows that the value of f at the center z_0 of the circle is the average of all the values of f on the circumference of C.

6 Series and Residues

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- **6.6.5** Summing Infinite Series

6.7 Applications
Chapter 6 Review Quiz



Special contour used in evaluating a real integral. See page 310.

In This Chapter Cauchy's integral formula for derivatives indicates that if a function f is analytic at a point z_0, then it possesses derivatives of all orders at that point. As a consequence of this result we shall see that f can always be expanded in a power series centered at that point. On the other hand, if f fails to be analytic at z_0, we may still be able to expand it in a different kind of series known as a *Laurent series.* The notion of Laurent series leads to the concept of a *residue*, and this, in turn, leads to yet another way of evaluating complex and, in some instances, real integrals.

6.1 Sequences and Series

INTRODUCTION Much of the theory of complex sequences and series is analogous to that encountered in real calculus. In this section we explore the definitions of convergence and divergence for complex sequences and complex infinite series. In addition, we give some tests for convergence of infinite series. You are urged to pay particular attention to what is said about *geometric series* since this type of series will be important in the later sections of this chapter.

A review of real sequences and series can be found in Section 6.1 of the student study guide. ▶

Sequences

A complex **sequence** $\{z_n\}$ is a function whose domain is the set of positive integers and whose range is a subset of the complex numbers $\mathbf{C}$. In other words, to each integer $n = 1, 2, 3, \ldots$ we assign a single complex number z_n. For example, the sequence $\{1 + i^n\}$ is

$$\begin{array}{ccccccc} 1+i, & 0, & 1-i, & 2, & 1+i, & \ldots. \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \\ n=1, & n=2, & n=3, & n=4, & n=5, & \ldots \end{array} \tag{1}$$

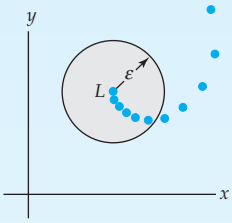


FIGURE 6.1.1 If $\{z_n\}$ converges to L, all but a finite number of terms are in every ε-neighborhood of L.

If $\lim_{n\to\infty} z_n = L$, then we say the sequence $\{z_n\}$ is **convergent.** In other words, $\{z_n\}$ converges to the number L if for each positive real number ε a positive integer N can be found such that $|z_n - L| < \varepsilon$ whenever $n > N$. Since $|z_n - L|$ is distance, the terms z_n of a sequence that converges to L can be made arbitrarily close to L. Put another way, when a sequence $\{z_n\}$ converges to L, then all but a finite number of the terms of the sequence are within every ε-neighborhood of L. See FIGURE 6.1.1. A sequence that is not convergent is said to be **divergent**.

The sequence $\{1 + i^n\}$ illustrated in (1) is divergent since the general term $z_n = 1 + i^n$ does not approach a fixed complex number as $n \to \infty$. Indeed, you should verify that the first four terms of this sequence repeat endlessly as n increases.

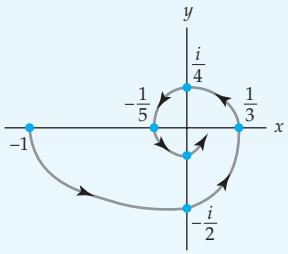


FIGURE 6.1.2 The terms of the sequence $\{i^{n+1}/n\}$ spiral in toward 0.

EXAMPLE 1 A Convergent Sequence

The sequence $\left\{\dfrac{i^{n+1}}{n}\right\}$ converges since $\lim_{n\to\infty} \dfrac{i^{n+1}}{n} = 0$. As we see from

$$-1, -\frac{i}{2}, \frac{1}{3}, \frac{i}{4}, -\frac{1}{5}, \cdots,$$

and FIGURE 6.1.2, the terms of the sequence, marked by colored dots in the figure, spiral in toward the point $z = 0$ as n increases. ☰

The following theorem for sequences is the analogue of Theorem 3.1.1 in Section 3.1.

THEOREM 6.1.1 Criterion for Convergence

A sequence $\{z_n\}$ converges to a complex number $L = a + ib$ if and only if $\text{Re}(z_n)$ converges to $\text{Re}(L) = a$ and $\text{Im}(z_n)$ converges to $\text{Im}(L) = b$.

EXAMPLE 2 Illustrating Theorem 6.1.1

Consider the sequence $\left\{\dfrac{3+ni}{n+2ni}\right\}$. From

$$z_n = \frac{3+ni}{n+2ni} = \frac{(3+ni)(n-2ni)}{n^2+4n^2} = \frac{2n^2+3n}{5n^2} + i\frac{n^2-6n}{5n^2},$$

we see that

$$\text{Re}(z_n) = \frac{2n^2+3n}{5n^2} = \frac{2}{5} + \frac{3}{5n} \to \frac{2}{5}$$

and

$$\text{Im}(z_n) = \frac{n^2-6n}{5n^2} = \frac{1}{5} - \frac{6}{5n} \to \frac{1}{5}$$

as $n \to \infty$. From Theorem 6.1.1, the last results are sufficient for us to conclude that the given sequence converges to $a + ib = \frac{2}{5} + \frac{1}{5}i$. ☰

Series An **infinite series** or **series** of complex numbers

$$\sum_{k=1}^{\infty} z_k = z_1 + z_2 + z_3 + \cdots + z_n + \cdots$$

is **convergent** if the sequence of partial sums $\{S_n\}$, where

$$S_n = z_1 + z_2 + z_3 + \cdots + z_n$$

converges. If $S_n \to L$ as $n \to \infty$, we say that the series converges to L or that the **sum** of the series is L.

Geometric Series A **geometric series** is any series of the form

$$\sum_{k=1}^{\infty} az^{k-1} = a + az + az^2 + \cdots + az^{n-1} + \cdots. \tag{2}$$

For (2), the nth term of the sequence of partial sums is

$$S_n = a + az + az^2 + \cdots + az^{n-1}. \tag{3}$$

When an infinite series is a geometric series, it is always possible to find a formula for S_n. To demonstrate why this is so, we multiply S_n in (3) by z,

$$zS_n = az + az^2 + az^3 + \cdots + az^n,$$

and subtract this result from S_n. All terms cancel except the first term in S_n and the last term in zS_n:

$$\begin{aligned} S_n - zS_n &= \left(a + az + az^2 + \cdots + az^{n-1}\right) \\ &\quad - \left(az + az^2 + az^3 + \cdots + az^{n-1} + az^n\right) \\ &= a - az^n \end{aligned}$$

or $(1-z)S_n = a(1-z^n)$. Solving the last equation for S_n gives us

$$S_n = \frac{a(1-z^n)}{1-z}. \tag{4}$$

Now $z^n \to 0$ as $n \to \infty$ whenever $|z| < 1$, and so $S_n \to a/(1-z)$. In other words, for $|z| < 1$ the sum of a geometric series (2) is $a/(1-z)$:

$$\frac{a}{1-z} = a + az + az^2 + \cdots + az^{n-1} + \cdots. \tag{5}$$

A geometric series (2) diverges when $|z| \geq 1$.

Special Geometric Series We next present several immediate deductions from (4) and (5) that will be particularly helpful in the two sections that follow. If we set $a = 1$, the equality in (5) is

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + \cdots. \tag{6}$$

You should remember (6) and (7). ▶

If we then replace the symbol z by $-z$ in (6), we get a similar result

$$\frac{1}{1+z} = 1 - z + z^2 - z^3 + \cdots. \tag{7}$$

Like (5), the equality in (7) is valid for $|z| < 1$ since $|-z| = |z|$. Now with $a = 1$, (4) gives us the sum of the first n terms of the series in (6):

$$\frac{1-z^n}{1-z} = 1 + z + z^2 + z^3 + \cdots + z^{n-1}.$$

If we rewrite the left side of the above equation as $\dfrac{1-z^n}{1-z} = \dfrac{1}{1-z} + \dfrac{-z^n}{1-z}$, we obtain an alternative form

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + \cdots + z^{n-1} + \frac{z^n}{1-z} \tag{8}$$

that will be put to use in proving the two principal theorems of this chapter.

EXAMPLE 3 Convergent Geometric Series

The infinite series

$$\sum_{k=1}^{\infty} \frac{(1+2i)^k}{5^k} = \frac{1+2i}{5} + \frac{(1+2i)^2}{5^2} + \frac{(1+2i)^3}{5^3} + \cdots$$

is a geometric series. It has the form given in (2) with $a = \frac{1}{5}(1+2i)$ and $z = \frac{1}{5}(1+2i)$. Since $|z| = \sqrt{5}/5 < 1$, the series is convergent and its sum is given by (5):

$$\sum_{k=1}^{\infty} \frac{(1+2i)^k}{5^k} = \frac{\dfrac{1+2i}{5}}{1 - \dfrac{1+2i}{5}} = \frac{1+2i}{4-2i} = \frac{1}{2}i. \qquad \equiv$$

We turn now to some important theorems about convergence and divergence of an infinite series. You should have seen similar theorems in a course in elementary calculus.

THEOREM 6.1.2 A Necessary Condition for Convergence

If $\sum_{k=1}^{\infty} z_k$ converges, then $\lim_{n\to\infty} z_n = 0$.

Proof Let L denote the sum of the series. Then $S_n \to L$ and $S_{n-1} \to L$ as $n \to \infty$. By taking the limit of both sides of $S_n - S_{n-1} = z_n$ as $n \to \infty$ we obtain the desired conclusion. ≡

A Test for Divergence The contrapositive* of the proposition in Theorem 6.1.2 is the familiar nth term test for divergence of an infinite series.

*See the footnote on page 132.

THEOREM 6.1.3 The *n*th Term Test for Divergence

If $\lim_{n\to\infty} z_n \neq 0$, then $\sum_{k=1}^{\infty} z_k$ diverges.

For example, the series $\sum_{k=1}^{\infty} (ik+5)/k$ diverges since $z_n = (in+5)/n \to i \neq 0$ as $n \to \infty$. The geometric series (2) diverges if $|z| \geq 1$ because even in the case when $\lim_{n\to\infty} |z^n|$ exists, the limit is not zero.

DEFINITION 6.1.1 Absolute and Conditional Convergence

An infinite series $\sum_{k=1}^{\infty} z_k$ is said to be **absolutely convergent** if $\sum_{k=1}^{\infty} |z_k|$ converges. An infinite series $\sum_{k=1}^{\infty} z_k$ is said to be **conditionally convergent** if it converges but $\sum_{k=1}^{\infty} |z_k|$ diverges.

In elementary calculus a real series of the form $\sum_{k=1}^{\infty} \frac{1}{k^p}$ is called a ***p*-series** and converges for $p > 1$ and diverges for $p \leq 1$. We use this well-known result in the next example.

EXAMPLE 4 Absolute Convergence

The series $\sum_{k=1}^{\infty} \frac{i^k}{k^2}$ is absolutely convergent since the series $\sum_{k=1}^{\infty} \left|\frac{i^k}{k^2}\right|$ is the same as the real convergent p-series $\sum_{k=1}^{\infty} \frac{1}{k^2}$. Here we identify $p = 2 > 1$. ☰

As in real calculus:

• *Absolute convergence implies convergence.*

See Problem 47 in Exercises 6.1. We are able to conclude that the series in Example 4,

$$\sum_{k=1}^{\infty} \frac{i^k}{k^2} = i - \frac{1}{2^2} - \frac{i}{3^2} + \cdots$$

converges because it is was shown to be absolutely convergent.

Tests for Convergence Two of the most frequently used tests for convergence of infinite series are given in the next theorems.

THEOREM 6.1.4 Ratio Test

Suppose $\sum_{k=1}^{\infty} z_k$ is a series of nonzero complex terms such that

$$\lim_{n\to\infty} \left|\frac{z_{n+1}}{z_n}\right| = L. \tag{9}$$

(*i*) If $L < 1$, then the series converges absolutely.

(*ii*) If $L > 1$ or $L = \infty$, then the series diverges.

(*iii*) If $L = 1$, the test is inconclusive.

THEOREM 6.1.5 Root Test

Suppose $\sum_{k=1}^{\infty} z_k$ is a series of complex terms such that

$$\lim_{n\to\infty} \sqrt[n]{|z_n|} = L. \tag{10}$$

(*i*) If $L < 1$, then the series converges absolutely.

(*ii*) If $L > 1$ or $L = \infty$, then the series diverges.

(*iii*) If $L = 1$, the test is inconclusive.

Proofs of the ratio and root tests are similar to those of their real counterparts. See Problem 48 in Exercises 6.1. We are interested primarily in applying the tests in Theorems 6.1.4 and 6.1.5 to power series.

Power Series The notion of a power series is important in the study of analytic functions. An infinite series of the form

$$\sum_{k=0}^{\infty} a_k(z - z_0)^k = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots, \tag{11}$$

where the coefficients a_k are complex constants, is called a **power series** in $z - z_0$. The power series (11) is said to be **centered at z_0**; the complex point z_0 is referred to as the **center** of the series. In (11) it is also convenient to define $(z - z_0)^0 = 1$ even when $z = z_0$.

Circle of Convergence Every complex power series (11) has a **radius of convergence**. Analogous to the concept of an interval of convergence for real power series, a complex power series (11) has a **circle of convergence**, which is the circle centered at z_0 of largest radius $R > 0$ for which (11) converges at every point *within* the circle $|z - z_0| = R$. A power series converges absolutely at all points z within its circle of convergence, that is, for all z satisfying $|z - z_0| < R$, and diverges at all points z exterior to the circle, that is, for all z satisfying $|z - z_0| > R$. The radius of convergence can be:

(*i*) $R = 0$ (in which case (11) converges only at its center $z = z_0$),

(*ii*) R a finite positive number (in which case (11) converges at all interior points of the circle $|z - z_0| = R$), or

(*iii*) $R = \infty$ (in which case (11) converges for all z).

A power series may converge at some, all, or at none of the points *on* the actual circle of convergence. See FIGURE 6.1.3 and the next example.

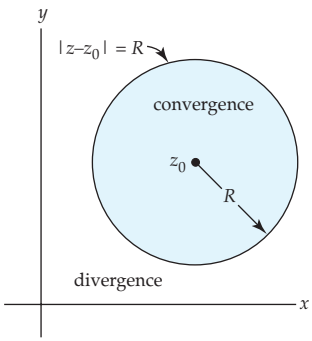


FIGURE 6.1.3 No general statement concerning convergence at points on the circle $|z - z_0| = R$ can be made.

EXAMPLE 5 Circle of Convergence

Consider the power series $\sum_{k=1}^{\infty} \frac{z^{k+1}}{k}$. By the ratio test (9),

$$\lim_{n\to\infty} \left| \frac{\frac{z^{n+2}}{n+1}}{\frac{z^{n+1}}{n}} \right| = \lim_{n\to\infty} \frac{n}{n+1}|z| = |z|.$$

Thus the series converges absolutely for $|z| < 1$. The circle of convergence is $|z| = 1$ and the radius of convergence is $R = 1$. Note that on the circle of convergence $|z| = 1$, the series does not converge absolutely since $\sum_{k=1}^{\infty} \frac{1}{k}$ is the well-known divergent harmonic series. Bear in mind *this does not say* that the series diverges on the circle of convergence. In fact, at $z = -1$, $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$ is the convergent alternating harmonic series. Indeed, it can be shown that the series converges at all points on the circle $|z| = 1$ *except* at $z = 1$.

It should be clear from Theorem 6.1.4 and Example 5 that for a power series $\sum_{k=0}^{\infty} a_k(z - z_0)^k$, the limit (9) depends only on the coefficients a_k. Thus, if

$$(i)\ \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L \neq 0, \text{ the radius of convergence is } R = \frac{1}{L}; \tag{12}$$

$$(ii)\ \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 0, \text{ the radius of convergence is } R = \infty; \tag{13}$$

$$(iii)\ \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \infty, \text{ the radius of convergence is } R = 0. \tag{14}$$

Similar conclusions can be made for the root test (10) by utilizing

$$\lim_{n\to\infty} \sqrt[n]{|a_n|}. \tag{15}$$

For example, if $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L \neq 0$, then $R = 1/L$.

EXAMPLE 6 Radius of Convergence

Consider the power series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k!}(z - 1 - i)^k$. With the identification $a_n = (-1)^{n+1}/n!$ we have

$$\lim_{n\to\infty}\left|\frac{\frac{(-1)^{n+2}}{(n+1)!}}{\frac{(-1)^{n+1}}{n!}}\right| = \lim_{n\to\infty} \frac{1}{n+1} = 0.$$

Hence by (13) the radius of convergence is ∞; the power series with center $z_0 = 1 + i$ converges absolutely for all z, that is, for $|z - 1 - i| < \infty$.

EXAMPLE 7 Radius of Convergence

Consider the power series $\sum_{k=1}^{\infty} \left(\frac{6k+1}{2k+5}\right)^k (z - 2i)^k$. With $a_n = \left(\frac{6n+1}{2n+5}\right)^n$, the root test in the form (15) gives

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{6n+1}{2n+5} = 3.$$

By reasoning similar to that leading to (12), we conclude that the radius of convergence of the series is $R = \frac{1}{3}$. The circle of convergence is $|z - 2i| = \frac{1}{3}$; the power series converges absolutely for $|z - 2i| < \frac{1}{3}$. ☰

The Arithmetic of Power Series On occasion it may be to our advantage to perform certain *arithmetic operations* on one or more power series. Although it would take us too far afield to delve into properties of power series in a formal manner (stating and proving theorems), it will be helpful at points in this chapter to know what we can (or cannot) do to power series. So here are some facts.

- A power series $\sum_{k=0}^{\infty} a_k(z - z_0)^k$ can be multiplied by a nonzero complex constant c without affecting its convergence or divergence.
- A power series $\sum_{k=0}^{\infty} a_k(z - z_0)^k$ converges absolutely within its circle of convergence. As a consequence, within the circle of convergence the terms of the series can be rearranged and the rearranged series has the same sum L as the original series.
- Two power series $\sum_{k=0}^{\infty} a_k(z - z_0)^k$ and $\sum_{k=0}^{\infty} b_k(z - z_0)^k$ can be added and subtracted by adding or subtracting like terms. In symbols:

$$\sum_{k=0}^{\infty} a_k(z - z_0)^k \pm \sum_{k=0}^{\infty} b_k(z - z_0)^k = \sum_{k=0}^{\infty} (a_k \pm b_k)(z - z_0)^k.$$

  If both series have the same nonzero radius R of convergence, the radius of convergence of $\sum_{k=0}^{\infty} (a_k \pm b_k)(z - z_0)^k$ is R. It should make intuitive sense that if one series has radius of convergence $r > 0$ and the other has radius of convergence $R > 0$, where $r \neq R$, then the radius of convergence of $\sum_{k=0}^{\infty} (a_k \pm b_k)(z - z_0)^k$ is the smaller of the two numbers r and R.
- Two power series can (with care) be multiplied and divided.

Remarks

(*i*) If $z_n = a_n + ib_n$ then the nth term of the sequence of partial sums for $\sum_{k=1}^{\infty} z_k$ can be written $S_n = \sum_{k=1}^{n} (a_k + ib_k) = \sum_{k=1}^{n} a_k + i\sum_{k=1}^{n} b_k$. Analogous to Theorem 6.1.1, $\sum_{k=1}^{\infty} z_k$ converges to a number $L = a + ib$ if and only if $\text{Re}(S_n) = \sum_{k=1}^{n} a_k$ converges to a and $\text{Im}(S_n) = \sum_{k=1}^{n} b_k$ converges to b. See Problem 35 in Exercises 6.1.

(*ii*) When written in terms of summation notation, a geometric series may not be immediately recognizable as equivalent to (2). In summation notation a geometric series need not start at $k = 1$ nor does the general term have to appear precisely as az^{k-1}. At first glance $\sum_{k=3}^{\infty} 40\,\frac{i^{k+2}}{2^{k-1}}$ does not appear to match the general form $\sum_{k=1}^{\infty} az^{k-1}$ of a geometric series. However, by writing out three terms,

$$\sum_{k=3}^{\infty} 40\frac{i^{k+2}}{2^{k-1}} = \overbrace{40\frac{i^5}{2^2}}^{a} + \overbrace{40\frac{i^6}{2^3}}^{az} + \overbrace{40\frac{i^7}{2^4}}^{az^2} + \cdots$$

we are able make the identifications $a = 40\left(i^5/2^2\right)$ and $z = i/2$ on the right-hand side of the equality. Since $|z| = \frac{1}{2} < 1$ the sum of the series is given by (5):

$$\sum_{k=3}^{\infty} 40\frac{i^{k+2}}{2^{k-1}} = \frac{40\dfrac{i^5}{2^2}}{1-\dfrac{i}{2}} = -4 + 8i.$$

(*iii*) Although we have not proved it, it bears repeating: A power series $\sum_{k=0}^{\infty} a_k(z - z_0)^k$, $z \neq z_0$, *always* possesses a radius of convergence R that is either positive, 0, or ∞. We have seen in the discussion prior to Example 6 that the ratio and root tests lead to

$$\frac{1}{R} = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| \quad \text{and} \quad \frac{1}{R} = \lim_{n\to\infty}\sqrt[n]{|a_n|}$$

assuming the appropriate limit exists. Since these formulas depend only on the coefficients, it is easy to make up examples where neither $\lim_{n\to\infty}|a_{n+1}/a_n|$ nor $\lim_{n\to\infty}|a_n|^{1/n}$ exist. What is R if neither of these limits exist? See Problems 45 and 46 in Exercises 6.1.

EXERCISES 6.1

Answers to selected odd-numbered problems begin on page ANS-16.

In Problems 1–4, write out the first five terms of the given sequence.

1. $\{5i^n\}$

2. $\{2 + (-i)^n\}$

3. $\left\{1 + e^{n\pi i}\right\}$

4. $\{(1 + i)^n\}$ [*Hint:* Write in polar form.]

In Problems 5–10, determine whether the given sequence converges or diverges.

5. $\left\{\dfrac{3ni + 2}{n + ni}\right\}$

6. $\left\{\dfrac{ni + 2^n}{3ni + 5^n}\right\}$

7. $\left\{\dfrac{(ni + 2)^2}{n^2 i}\right\}$

8. $\left\{\dfrac{n(1 + i^n)}{n + 1}\right\}$

9. $\left\{\dfrac{n + i^n}{\sqrt{n}}\right\}$

10. $\left\{e^{1/n} + 2\,(\tan^{-1} n)i\right\}$

In Problems 11 and 12, show that the given sequence $\{z_n\}$ converges to a complex number L by computing $\lim_{n\to\infty}\mathrm{Re}(z_n)$ and $\lim_{n\to\infty}\mathrm{Im}(z_n)$.

11. $\left\{\dfrac{4n + 3ni}{2n + i}\right\}$

12. $\left\{\left(\dfrac{1 + i}{4}\right)^n\right\}$

In Problems 13 and 14, use the sequence of partial sums to show that the given series is convergent.

13. $\displaystyle\sum_{k=1}^{\infty}\left[\frac{1}{k + 2i} - \frac{1}{k + 1 + 2i}\right]$

14. $\displaystyle\sum_{k=1}^{\infty}\frac{i}{k(k + 1)}$

In Problems 15–20, determine whether the given geometric series is convergent or divergent. If convergent, find its sum.

15. $\displaystyle\sum_{k=0}^{\infty}(1 - i)^k$

16. $\displaystyle\sum_{k=1}^{\infty} 4i\left(\tfrac{1}{3}\right)^{k-1}$

17. $\displaystyle\sum_{k=1}^{\infty}\left(\frac{i}{2}\right)^k$

18. $\displaystyle\sum_{k=0}^{\infty}\tfrac{1}{2}i^k$

19. $\displaystyle\sum_{k=0}^{\infty} 3\left(\frac{2}{1 + 2i}\right)^k$

20. $\displaystyle\sum_{k=2}^{\infty}\frac{i^k}{(1 + i)^{k-1}}$

In Problems 21–30, find the circle and radius of convergence of the given power series.

21. $\sum_{k=0}^{\infty} \frac{1}{(1-2i)^{k+1}}(z-2i)^k$

22. $\sum_{k=1}^{\infty} \frac{1}{k}\left(\frac{i}{1+i}\right) z^k$

23. $\sum_{k=1}^{\infty} \frac{(-1)^k}{k2^k}(z-1-i)^k$

24. $\sum_{k=1}^{\infty} \frac{1}{k^2(3+4i)^k}(z+3i)^k$

25. $\sum_{k=0}^{\infty} (1+3i)^k(z-i)^k$

26. $\sum_{k=1}^{\infty} \frac{z^k}{k^k}$

27. $\sum_{k=0}^{\infty} \frac{(z-4-3i)^k}{5^{2k}}$

28. $\sum_{k=0}^{\infty} (-1)^k \left(\frac{1+2i}{2}\right)^k (z+2i)^k$

29. $\sum_{k=0}^{\infty} \frac{(2k)!}{(k+2)\,(k!)^2}(z-i)^{2k}$

30. $\sum_{k=0}^{\infty} \frac{k!}{(2k)^k} z^{3k}$

31. Show that the power series $\sum_{k=1}^{\infty} \frac{(z-i)^k}{k2^k}$ is not absolutely convergent on its circle of convergence. Determine at least one point on the circle of convergence at which the power series is convergent.

32. Show that the power series $\sum_{k=1}^{\infty} \frac{z^k}{k^2}$ converges at every point on its circle of convergence.

33. Reread Theorem 6.1.3. What conclusion can be drawn, if any, when $\lim_{n\to\infty} |z_n| \neq 0$?

34. Show that the power series $\sum_{k=1}^{\infty} kz^k$ diverges at every point on its circle of convergence.

Focus on Concepts

35. By considering the series $\sum_{k=0}^{\infty} r^k e^{ik\theta}$, $0 < r < 1$, show that

$$\sum_{k=0}^{\infty} r^k \cos k\theta = \frac{1 - r\cos\theta}{1 - 2r\cos\theta + r^2} \quad \text{and} \quad \sum_{k=0}^{\infty} r^k \sin k\theta = \frac{r\sin\theta}{1 - 2r\cos\theta + r^2}.$$

36. Suppose $\{z_n + w_n\}$ converges. Discuss: Does it follow that at least one of the sequences $\{z_n\}$ or $\{w_n\}$ converges?

37. A sequence $\{z_n\}$ is said to be **bounded** if the set S consisting of its terms is a bounded set (see Section 1.5). The sequence in Example 1 is bounded.

(**a**) Prove that the sequence in Example 2 is bounded.

(**b**) Give another example of a sequence consisting of complex terms that is bounded.

(**c**) Give an example of a sequence consisting of complex terms that is unbounded.

38. Discuss: Is every convergent sequence $\{z_n\}$ bounded? (See Problem 37.)
Is every bounded sequence convergent? Defend your answers with sound mathematics.

39. Does the sequence $\{i^{1/n}\}$, where $i^{1/n}$ denotes the principal nth root of i, converge?

40. We saw that the equality in (6) was valid for $|z| < 1$. Show that

$$\frac{1}{1-z} = -z^{-1} - z^{-2} - z^{-3} - \cdots$$

and give the values of z for which the equality is valid.

41. Consider (6) with the symbol z replaced by e^{iz}:

$$\frac{1}{1-e^{iz}} = 1 + e^{iz} + e^{2iz} + \cdots.$$

Give the region in the complex plane for which the foregoing series converges.

42. Sketch the region in the complex plane for which $\sum_{k=0}^{\infty}\left(\frac{z-1}{z+2}\right)^k$ converges.

43. Consider the power series $\sum_{k=0}^{\infty} a_k(z-1+2i)^k$. Discuss: Can the series converge at $-3+i$ and diverge at $5-3i$?

44. Use a sketch in the complex plane that illustrates the validity of each of the following theorems:

(*i*) If a power series centered at z_0 converges at $z_1 \neq z_0$, then the series converges for every z for which $|z-z_0| < |z_1-z_0|$.

(*ii*) If a power series centered at z_0 diverges at z_2, then the series diverges for every z for which $|z-z_0| > |z_2-z_0|$.

45. Consider the power series $f(z) = \sum_{k=0}^{\infty} a_k z^k$ where

$$a_k = \begin{cases} 2^k, & k = 0, 2, 4, \ldots \\ \dfrac{1}{7^k} & k = 1, 3, 5, \ldots \end{cases}.$$

(**a**) Show that neither $\lim_{n\to\infty}\left|a_{n+1}\big/a_n\right|$ nor $\lim_{n\to\infty}|a_n|^{1/n}$ exist.

(**b**) Find the radius of convergence of each power series:

$$f_1(z) = \sum_{k=0}^{\infty} 2^{2k} z^{2k} \qquad \text{and} \qquad f_2(z) = \sum_{k=0}^{\infty} \frac{1}{7^{2k+1}} z^{2k+1}.$$

(**c**) Verify that $f(z) = f_1(z) + f_2(z)$. Discuss: How can the radius of convergence R for the original power series be found from the foregoing observation? What is R?

46. Proceed as in Problem 45 to find the radius of convergence R for the power series

$$1 + 3z + z^2 + 27z^3 + z^4 + 243z^5 + z^6 + \cdots .$$

47. In this problem you are guided through the proof of the proposition:

If the series $\sum_{k=1}^{\infty} z_k$ converges absolutely, then the series converges.

Proof We begin with the hypothesis that $\sum_{k=1}^{\infty}|z_k|$ converges. If $z_k = a_k + ib_k$, then $\sum_{k=1}^{\infty}|a_k| \le \sum_{k=1}^{\infty}\sqrt{a_k^2+b_k^2} = \sum_{k=1}^{\infty}|z_k|$.

(**a**) First, explain why the foregoing inequality is true. Second, explain why this inequality shows that the series $\sum_{k=1}^{\infty}|a_k|$ converges.

(**b**) Explain how your reasoning in part (a) also shows that $\sum_{k=1}^{\infty}|b_k|$ converges.

(**c**) Explain how parts (a) and (b) show that $\sum_{k=1}^{\infty} z_k$ converges.

48. In this problem you are guided through the proof of part (*i*) of the ratio test (9).

Proof We begin with the hypothesis that $\lim_{n\to\infty}|z_{n+1}/z_n| = L < 1$.

(**a**) First, explain why there is an integer N and a positive real number $r < 1$ such that $|z_{n+1}/z_n| < r$ whenever $n > N$. Second, explain why this implies that $|z_{N+j}| < r^j|z_N|$ for $j = 0, 1, 2, \ldots$.

(**b**) Use the comparison test from real analysis and part (a) to show that $\sum_{j=0}^{\infty}|z_{N+j}|$ converges.

(**c**) Explain why part (b) shows that $\sum_{k=0}^{\infty} z_k$ converges absolutely.